Appendix Cancer Prediction Dataset - Function Analysis Report

Source Dataset: ankushpanday1/appendix-cancer-prediction-dataset | **Total Records:** 260,000 | **Total Columns:** 88 (25 Original + 63 Calculated) | **Generated:** 2026-09-01 11:31:31

1. Mathematical & Numeric Functions

Function Signature	Expression / Applied To	Original Input	Computed Output	Data Type
`ABS(number)`	ABS(BMI - 25.0)	BMI = 28.6	3.60	`float64`
`ROUND(number, decimals)`	ROUND(Survival_Years, 1)	Survival = 6.5	6.5	`float64`
`SQRT(number)`	SQRT(Age)	Age = 67	8.1854	`float64`
`POWER(number, power)`	POWER(BMI, 2)	BMI = 28.6	817.96	`float64`
`RAND()`	RAND()	[Random Generation]	0.204932	`float64`
`INT(expression)`	INT(BMI)	BMI = 28.6	28	`int64`
`FLOAT(expression)`	FLOAT(Blood_Pressure)	BP = 156	156.0	`float64`

2. String & Text Functions

Function Signature	Expression / Applied To	Original Input	Computed Output	Data Type
`UPPER(string)`	UPPER(Country)	Country = China	CHINA	`object`
`LOWER(string)`	LOWER(Diet_Type)	Diet = Non-Vegetarian	non-vegetarian	`object`
`LEN(string)`	LEN(Patient_ID)	ID = PID_000001	10	`int64`
`SPLIT(string, delim, tok)`	SPLIT(Patient_ID, '_', 2)	ID = PID_000001	000001	`object`
`LEFT(string, number)`	LEFT(Patient_ID, 3)	ID = PID_000001	PID	`object`
`TRIM(string)`	TRIM(Country)	Country = China	China	`object`
`MID(string, start, len)`	MID(Patient_ID, 5, 3)	ID = PID_000001	000	`object`
`RIGHT(string, number)`	RIGHT(Patient_ID, 4)	ID = PID_000001	0001	`object`
`REPLACE(string, sub, rep)`	REPLACE(Patient_ID, 'PID_')	ID = PID_000001	PATIENT_000001	`object`
`CONTAINS(string, sub)`	CONTAINS(Diet_Type, 'Veg')	Diet = Non-Vegetarian	True	`bool`
`STARTSWITH(string, sub)`	STARTSWITH(Patient_ID, 'PID')	ID = PID_000001	True	`bool`

3. Date & Time Functions

Function Signature	Expression / Applied To	Original Input	Computed Output	Data Type
`TODAY()`	TODAY()	[System Date]	2026-09-01	`date`
`NOW()`	NOW()	[System Timestamp]	2026-09-01 11:31	`datetime`
`MAKEDATE(year, month, day)`	MAKEDATE(2022, 8, 15)	Age = 67	2022-08-15	`date`
`DATEPART(part, date)`	DATEPART('year', Diag_Date)	Date = 2024-07-06	2024	`int32`
`DATENAME(part, date)`	DATENAME('month', Diag_Date)	Date = 2024-07-06	July	`object`
`DATEADD(part, num, date)`	DATEADD('day', Survival, Date)	Delay = 691d	2031-01-03	`datetime64`
`DATEDIFF(part, start, end)`	DATEDIFF('day', Date, TODAY)	Date = 2024-07-06	787 days	`int64`
`DATETRUNC(part, date)`	DATETRUNC('month', Diag_Date)	Date = 2024-07-06	2024-07-01	`datetime64`

4. Type & Logical Conversion Functions

Function Signature	Expression / Applied To	Original Input	Computed Output	Data Type
`STR(expression)`	STR(Age)	Age = 67 (int)	'67' (str)	`object`
`BOOL(expression)`	BOOL(Prediction == 'Yes')	Prediction = Yes	True	`bool`
`DATE(expression)`	DATE(Diagnosis_Date)	Timestamp	2024-07-06	`date`

5. Aggregation & Summary Functions

Function Signature	Expression / Applied To	Original Input	Computed Output	Data Type
`SUM(expression)`	SUM(Blood_Pressure)	Total Rows = 260,000	34,967,489	`int64`
`AVG(expression)`	AVG(BMI)	Dataset BMI Series	25.00	`float64`
`MAX(expression)`	MAX(Age)	Age Range	89 years	`int64`
`MIN(expression)`	MIN(Age)	Age Range	18 years	`int64`
`COUNT(expression)`	COUNT(Patient_ID)	Patient ID Series	260,000 patients	`int64`
`COUNTD(expression)`	COUNTD(Country)	Country Series	25 unique countries	`int64`

6. Logical & Conditional Functions

Function Signature	Expression / Applied To	Original Input	Computed Output	Data Type
`IF..THEN..ELSE..END`	IF Age>=60 THEN 'Senior'...	Age = 67	Senior	`object`
`IIF(cond, true, false)`	IIF(BMI>30 | Smoker, High, Std)	BMI=28.6, Smoke=No	Standard Risk	`object`
`AND Function`	AND(Age>=60, Smoking=='Yes')	Age=67, Smoke=No	False	`bool`
`OR Function`	OR(History=='Yes', Mutation=='Yes')	Hist=Yes, Mut=No	True	`bool`
`NOT Function`	NOT(Smoking == 'Yes')	Smoke = No	True	`bool`

7. Table Calculations & Window Functions

Function Signature	Expression / Applied To	Original Input	Computed Output	Data Type
`INDEX()`	INDEX()	Row Position	1	`int64`
`FIRST()`	FIRST()	Offset to 1st row	0	`int64`
`LAST()`	LAST()	Offset to last row	259999	`int64`
`LOOKUP(col, offset)`	LOOKUP(BMI, -1)	Prev Row BMI	28.6	`float64`
`RUNNING_SUM(col)`	RUNNING_SUM(Blood_Pressure)	Cumulative BP	156	`int64`
`WINDOW_SUM(col)`	WINDOW_SUM(Blood_Pressure)	Total Window BP	34,967,489	`int64`
`RANK(col)`	RANK(BMI, desc)	BMI = 28.6	Rank #60283	`int64`
`TOTAL(col)`	TOTAL(Blood_Pressure)	Dataset BP Total	34,967,489	`int64`

8. User Session & Security Functions

Function Signature	Expression / Applied To	Original Input	Computed Output	Data Type
`USERNAME()`	USERNAME()	[System User]	abhinav	`str`
`FULLNAME()`	FULLNAME()	[Full Name]	Abhinav Analyst	`str`
`ISFULLNAME()`	ISFULLNAME(Fullname)	[User Check]	True	`bool`
`ISMEMBEROF(group)`	ISMEMBEROF('Cancer_Research')	[Group Security]	True	`bool`
`USERDOMAIN()`	USERDOMAIN()	[Network Domain]	mitwpu.edu	`str`

9. Spatial & Geographic Functions

Function Signature	Expression / Applied To	Original Input	Computed Output	Data Type
`MAKEPOINT(lat, lon)`	MAKEPOINT(Lat, Lon)	Country = China	35.8617, 104.1954	`str`
`DISTANCE(pt1, pt2)`	DISTANCE(Patient, Hospital)	Country = China	11,498.4 km	`float64`
`BUFFER(pt, radius)`	BUFFER(Patient, 50km)	Lat/Lon	BUFFER(35.8617, 104.1954, 50.0km)	`str`
`INTERSECTS(g1, g2)`	INTERSECTS(Country, 'China')	Country = China	True	`bool`
`MAKELINE(pt1, pt2)`	MAKELINE(Patient, Hospital)	Lat/Lon -> Hospital	LINE((35.8617, 104.1954) -> (4...	`str`
`AREA(radius_km)`	AREA(50km)	Radius = 50 km	7,853.98 sq km	`float64`
`CENTROID(pt1, pt2)`	CENTROID(Patient, Hospital)	Midpoint	38.28725, 15.094700000000...	`str`

10. Predictive & Statistical Analytics Functions

Function Signature	Expression / Applied To	Original Input	Computed Output	Data Type
`MODEL_QUANTILE(col, q)`	MODEL_QUANTILE(BMI, 0.5)	BMI Median	25.00	`float64`
`MODEL_PERCENTILE(col)`	MODEL_PERCENTILE(BMI)	BMI = 28.6	Top 76.5%	`float64`
`FORECAST(col, periods)`	FORECAST(Survival, 1)	Survival History	4.88 yrs	`float64`
`TRENDLINE(x, y)`	TRENDLINE(Age, Survival)	Age vs Survival	Slope: 0.0148	`float64`

Sample Dataset Preview (First 5 Patients x Selected Key Calculated Fields)

Patient ID	Country	Age	BMI	BMI Dev.	Age Group	Risk Category	Rank	Distance (KM)
`PID_000001`	China	67	28.6	3.60	Senior	Standard Risk	`#60283`	11,498.4
`PID_000002`	Saudi Arabia	83	19.4	5.60	Senior	Standard Risk	`#225292`	10,471.7
`PID_000003`	France	52	17.2	7.80	Adult	Standard Risk	`#244200`	5,940.4
`PID_000004`	Russia	71	31.8	6.80	Senior	High Risk	`#22153`	8,646.7
`PID_000005`	USA	19	17.7	7.30	Youth	Standard Risk	`#240901`	1,915.8